

PH 101

Lecture 12

OUTLINE

- Ignorable coordinates
- Conservation Laws

Generalized momenta

In the last class we learnt about the so-called generalized momentum.

Consider a holonomic mechanical system with Lagrangian $L = L(\mathbf{q}, \dot{\mathbf{q}}, t)$. Then the quantity p_j defined by

$$p_j = \frac{\partial L}{\partial \dot{q}_j}$$

*is called the **generalised momentum** corresponding to the coordinate q_j . It is also called the momentum **conjugate** to q_j .*

Ignorable coordinates

When the Lagrangian is independent of a coordinate q_i the corresponding generalized momentum p_i is conserved.

When the Lagrangian is independent of a coordinate q_i , that coordinate is sometimes said to be **ignorable** or **cyclic**.

Obviously it is a good idea, when possible, to choose coordinates so that as many as possible are ignorable and their corresponding momenta are constant. In fact, this is perhaps the main criterion in choosing generalized coordinates for any given problem: Try to find coordinates as many as possible of which are ignorable.

“ L is independent of a coordinate q_i ” is equivalent to saying
" L , is unchanged, or invariant, when q_i varies (with all the other q_i held fixed)."

Thus we can say that if L is invariant under variations of a coordinate q_i then the corresponding generalized momentum p_i is conserved.

To re-emphasis again:

In terms of the generalised momentum p_j , the j -th Lagrange equation can be written

$$\frac{dp_j}{dt} = \frac{\partial L}{\partial q_j}.$$

It follows that if $\partial L / \partial q_j = 0$ (that is, if the coordinate q_j is absent from the Lagrangian), then the generalised momentum p_j is constant in any motion. Such ‘absent’ coordinates are said to be **cyclic**. We have therefore shown that:

If q_j is a cyclic coordinate (in the sense that it does not appear in the Lagrangian), then p_j , the generalised momentum conjugate to q_j , is constant in any motion.

Examples

Example 1:

Consider a system described by the Lagrangian:

$$L = \frac{1}{2}M\dot{x}^2 + \frac{1}{2}m \left(\dot{x}^2 + \dot{y}^2 + 2\dot{x}\dot{y} \cos \alpha \right) + mgy \sin \alpha.$$

This Lagrangian does not contain the coordinate x explicitly. L is independent of x . So, x is cyclic or ignorable coordinate, but not y .

The generalized momenta, p_x , corresponding to x is conserved.

$$p_x = \frac{\partial L}{\partial \dot{x}} = M\dot{x} + m(\dot{x} + \dot{y} \cos \alpha) \quad \text{is conserved}$$
$$p_y = \frac{\partial L}{\partial \dot{y}} = m(\dot{y} + \dot{x} \cos \alpha) \quad \text{is NOT conserved}$$

Example 2:

Consider a system described by the Lagrangian:

$$L = \frac{1}{2}ma^2 \left[\dot{\theta}^2 + (\sin \theta \dot{\phi})^2 \right] + mga \cos \theta.$$

This Lagrangian does not contain the coordinate ϕ explicitly. L is independent of ϕ . So, ϕ is **cyclic** or **ignorable** coordinate, but not θ . The generalized momenta, p_ϕ , corresponding to ϕ is conserved.

Since $\partial L / \partial \phi = 0$, the coordinate ϕ is **cyclic**. It follows that the conjugate momentum p_ϕ is conserved, where

$$p_\phi = \frac{\partial L}{\partial \dot{\phi}} = ma^2 \sin^2 \theta \dot{\phi}.$$

Conservation Laws from a Lagrangian perspective

Conservation of Total Momentum

We already know from Newtonian mechanics that the total momentum of an isolated system of N particles is conserved, but let us examine this important property from the Lagrangian point of view. One of the most prominent features of an isolated system is that it is translationally invariant; that is, if we transport all N particles bodily through the same displacement $\boldsymbol{\epsilon}$, nothing physically significant about the system should change. This is illustrated in Figure 1, where we see that the effect of moving the whole system through the fixed displacement $\boldsymbol{\epsilon}$ is to replace every position $\mathbf{r}_\alpha$ by $\mathbf{r}_\alpha + \boldsymbol{\epsilon}$,

$$\mathbf{r}_1 \rightarrow \mathbf{r}_1 + \boldsymbol{\epsilon}, \quad \mathbf{r}_2 \rightarrow \mathbf{r}_2 + \boldsymbol{\epsilon}, \quad \cdots, \quad \mathbf{r}_N \rightarrow \mathbf{r}_N + \boldsymbol{\epsilon}. \quad (1)$$

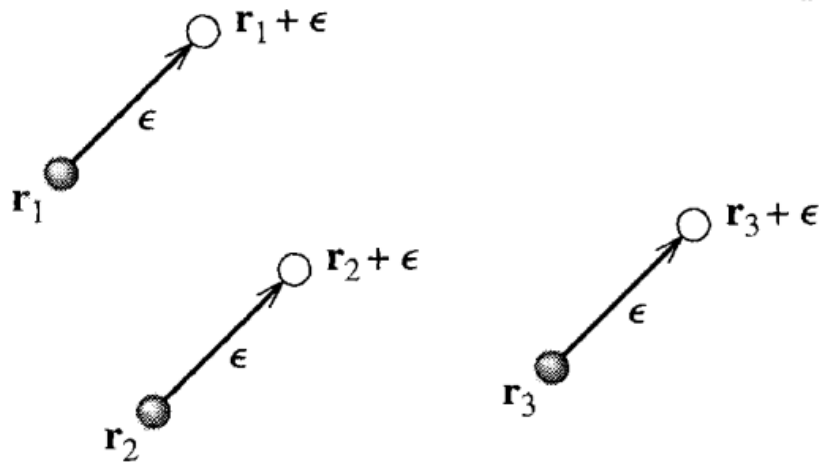


Fig. 1

In particular, the potential energy must be unaffected by this displacement, so that

$$V(\mathbf{r}_1 + \boldsymbol{\epsilon}, \cdots, \mathbf{r}_N + \boldsymbol{\epsilon}, t) = V(\mathbf{r}_1, \cdots, \mathbf{r}_N, t)$$

Or, more briefly: $\delta V = 0$

Under translation: $\delta V = 0$

where δV denotes the change in V under the translation (1). Clearly the velocities are unchanged by the translation (1). (Adding a constant ϵ to all the $\mathbf{r}_\alpha$, does not change the $\dot{\mathbf{r}}_\alpha$.) Therefore $\delta T = 0$, and hence

$$\delta L = 0$$

under the translation (1). This result is true for any displacement ϵ . If we choose ϵ to be an infinitesimal displacement in the $\mathbf{x}$ direction, then all of the $\mathbf{x}$ coordinates $x_1, x_2, \dots, x_N$ increase by ϵ , while the y and z coordinates are unchanged. For this translation, the change in L is

$$\delta L = \epsilon \frac{\partial L}{\partial x_1} + \epsilon \frac{\partial L}{\partial x_2} + \dots + \epsilon \frac{\partial L}{\partial x_N} = 0$$

This implies that

$$\sum_{\alpha=1}^N \frac{\partial L}{\partial x_\alpha} = 0$$

$$\sum_{\alpha=1}^N \frac{\partial L}{\partial x_{\alpha}} = 0 \quad (2)$$

Now using Lagrange's equations we can rewrite each derivative as

$$\frac{\partial L}{\partial x_{\alpha}} = \frac{d}{dt} \frac{\partial L}{\partial \dot{x}_{\alpha}} = \frac{d}{dt} p_{\alpha x}$$

where $p_{\alpha x}$ is the x component of the momentum of particle α .

Thus (2) becomes

$$\sum_{\alpha=1}^N \frac{d}{dt} p_{\alpha x} = \frac{d}{dt} P_x = 0$$

where P_x is the x component of the total momentum $\mathbf{P} = \sum_{\alpha} \mathbf{p}_{\alpha}$. By choosing the small displacement ϵ successively in the y and z directions, we can prove the same result for the y and z components, and we reach the conclusion that, provided the Lagrangian is unchanged by the translation (1), the total momentum of the N-particle system is conserved.

The **conservation law of momentum** follows from the **homogeneity of space**. By virtue of this homogeneity, the mechanical properties of a closed system are unchanged by any parallel displacement of the entire system in space. A parallel displacement is a transformation in which every particle in the system is moved by the same amount, the radius vector $\mathbf{r}$ becoming $\mathbf{r} + \mathbf{\epsilon}$, as we have already discussed.

Conservation of Energy

Let us consider the conservation law resulting from the **homogeneity of time**. By virtue of this homogeneity, the Lagrangian of a closed system does not depend explicitly on time. The total time derivative of the Lagrangian can therefore be written

$$\frac{dL}{dt} = \sum_i \frac{\partial L}{\partial q_i} \dot{q}_i + \sum_i \frac{\partial L}{\partial \dot{q}_i} \ddot{q}_i$$

If L depended explicitly on time, a term $\frac{\partial L}{\partial t}$ would have to be added on the right-hand side. Replacing $\frac{\partial L}{\partial q_i}$ in accordance with Lagrange's equations, by $\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i}$, we obtain

$$\begin{aligned} \frac{dL}{dt} &= \sum_i \dot{q}_i \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) + \sum_i \frac{\partial L}{\partial \dot{q}_i} \ddot{q}_i \\ &= \sum_i \frac{d}{dt} \left(\dot{q}_i \frac{\partial L}{\partial \dot{q}_i} \right) \end{aligned}$$

$$\begin{aligned}\frac{dL}{dt} &= \sum_i \dot{q}_i \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) + \sum_i \frac{\partial L}{\partial \dot{q}_i} \ddot{q}_i \\ &= \sum_i \frac{d}{dt} \left(\dot{q}_i \frac{\partial L}{\partial \dot{q}_i} \right)\end{aligned}$$

Or

$$\frac{d}{dt} \left(\sum_i \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} - L \right) = 0$$

Hence we see that the quantity

$$E \equiv \sum_i \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} - L$$

remains constant during the motion of a closed system, i.e. it is an integral of the motion; it is called the **energy** of the system.

Important Facts about E :

When could it be termed as energy?

Theorem *A necessary and sufficient condition for the quantity E to be the total energy of a system whose Lagrangian is written in terms of a set of coordinates q_i is that these q_i are related to a Cartesian set of coordinates x_i via expressions of the form,*

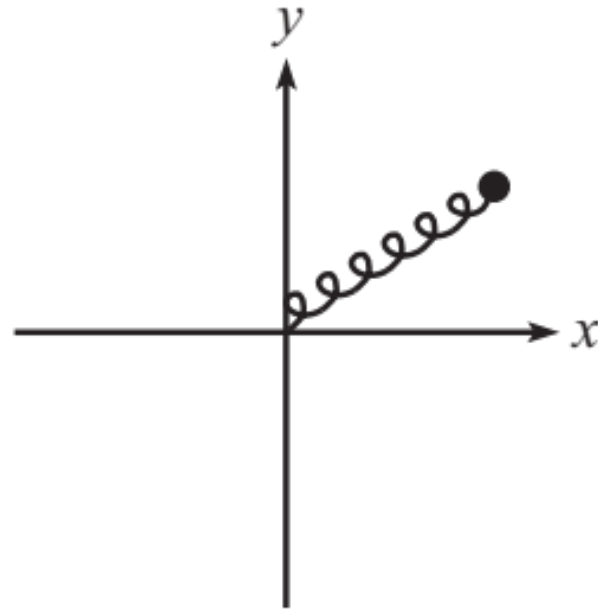
$$\begin{aligned}x_1 &= x_1(q_1, q_2, \dots), \\ &\vdots \\ x_N &= x_N(q_1, q_2, \dots).\end{aligned}$$

That is, there is no t or $\dot{q}_i$ dependence.

The law of conservation of energy is valid not only for closed systems, but also for those in a constant external field (i.e. one independent of time): **the only property of the Lagrangian used in the above derivation, namely that it does not involve the time explicitly, is still valid.** Mechanical systems whose energy is conserved are sometimes called conservative systems.

Examples

Example 1 (Particle in a plane): A particle of mass m moves in a horizontal plane. It is connected to the origin by a spring with spring constant k and relaxed length zero (so the potential energy is $kr^2/2 = k(x^2 + y^2)/2$), as shown in Fig. Find L and E in terms of Cartesian coordinates, and then also in terms of polar coordinates. Verify that in both cases, E is the energy and it is conserved.



Solution: In Cartesian coordinates, we have

$$L = T - V = \frac{m}{2}(\dot{x}^2 + \dot{y}^2) - \frac{k}{2}(x^2 + y^2),$$

and so

$$E = \frac{\partial L}{\partial \dot{x}}\dot{x} + \frac{\partial L}{\partial \dot{y}}\dot{y} - L = \frac{m}{2}(\dot{x}^2 + \dot{y}^2) + \frac{k}{2}(x^2 + y^2),$$

which is indeed the energy.

In polar coordinates, we have

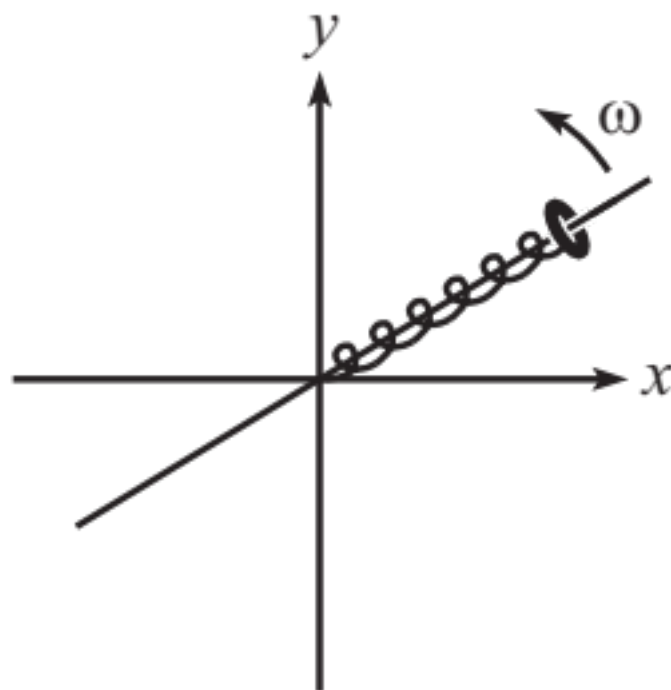
$$L = T - V = \frac{m}{2}(\dot{r}^2 + r^2\dot{\theta}^2) - \frac{kr^2}{2},$$

and so

$$E = \frac{\partial L}{\partial \dot{r}}\dot{r} + \frac{\partial L}{\partial \dot{\theta}}\dot{\theta} - L = \frac{m}{2}(\dot{r}^2 + r^2\dot{\theta}^2) + \frac{kr^2}{2},$$

which is again the energy. The Cartesian-coordinate E is always the energy. In both cases, E is conserved because L has no explicit t dependence

Example 2 (Bead on a rod): A bead of mass m is constrained to move along a massless rod that is pivoted at the origin and arranged (via an external torque) to rotate with constant angular speed ω in a horizontal plane. A spring with spring constant k and relaxed length zero lies along the rod and connects the mass to the origin, as shown in Fig. Find L and E in terms of the polar coordinate r , and show that E is conserved but it is not the energy.



Solution: The kinetic energy comes from both radial and tangential motion, so the Lagrangian is

$$L = \frac{m}{2}(\dot{r}^2 + r^2\omega^2) - \frac{kr^2}{2}.$$

E is then

$$E = \frac{\partial L}{\partial \dot{r}}\dot{r} - L = \frac{m}{2}(\dot{r}^2 - r^2\omega^2) + \frac{kr^2}{2}.$$

Since L has no explicit t dependence, so E is conserved. However, E is not the energy, due to the minus sign in front of the $r^2\omega^2$ term. This is consistent with the **Theorem**, because the relations between the Cartesian coordinates and the coordinate r (namely $x = r \cos \omega t$ and $y = r \sin \omega t$) involve t and are therefore not of the form of Equations:

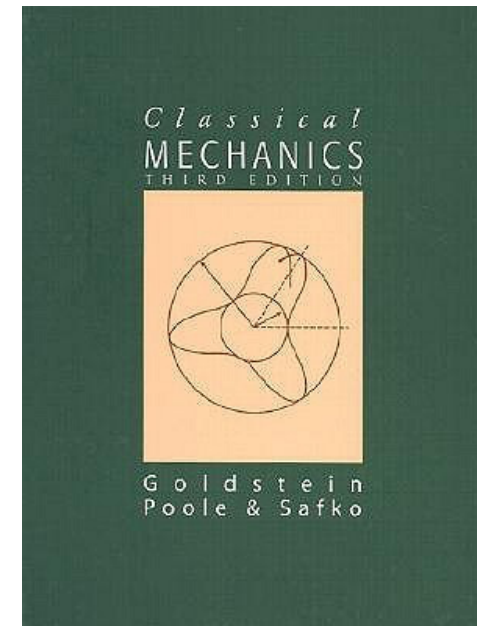
$$\begin{aligned}x_1 &= x_1(q_1, q_2, \dots), \\ &\vdots \\ x_N &= x_N(q_1, q_2, \dots).\end{aligned}$$

Conservation of Angular Momentum

This conservation law follows from the **isotropy of space**. This isotropy means that the mechanical properties of a closed system do not vary when it is rotated as a whole in any manner in space.

*We are not going to discuss it in this course. Interested students can study it in **Classical Mechanics** by H. Goldstein.*

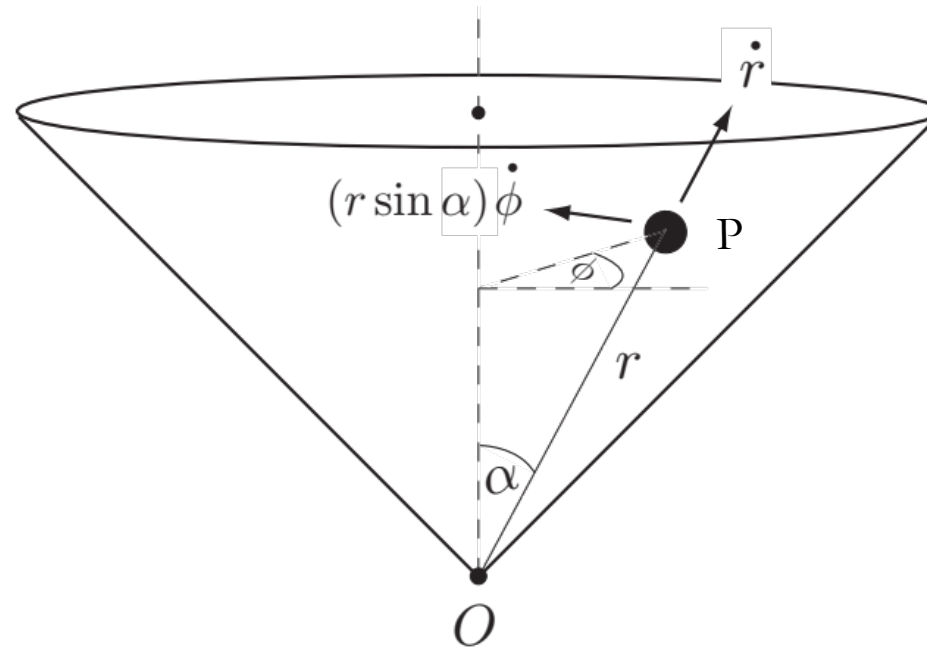
This topic on conservation of angular momentum from Lagrangian perspective is not going to be asked in examination!



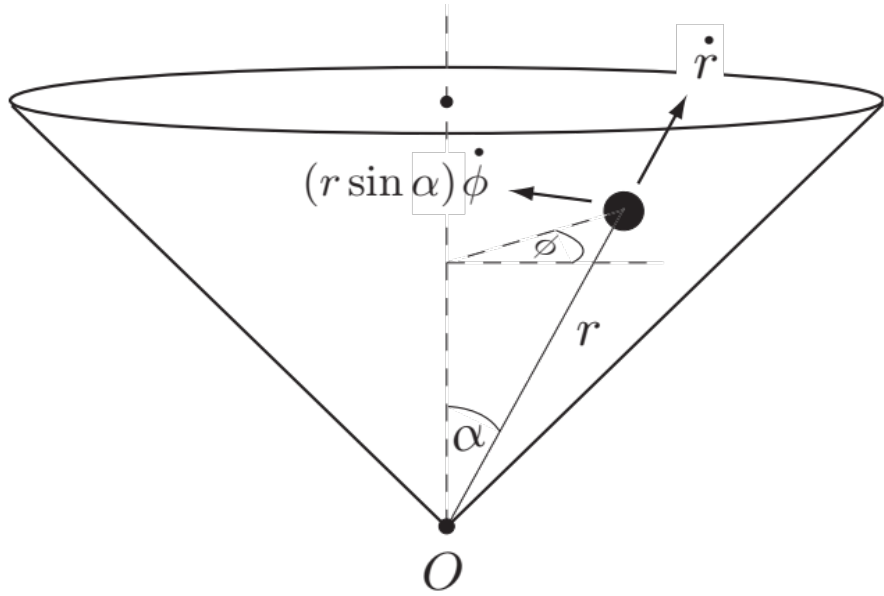
More Examples

Example 1

A particle P of mass m slides on the smooth inner surface of a circular cone of semi-angle α . The axis of symmetry of the cone is vertical with the vertex O pointing downwards. Take as generalised coordinates r , the distance OP , and ϕ , the azimuthal angle about the vertical through O . Obtain Lagrange's equations. Show that ϕ is a cyclic coordinate and find (and identify) the conserved momentum p_ϕ .



Solution:



The **kinetic energy** of the particle is

$$T = \frac{1}{2}m \left(\dot{r}^2 + r^2 \sin^2 \alpha \dot{\phi}^2 \right)$$

and the **potential energy** of the particle (relative to O) is

$$V = mgr \cos \alpha.$$

The **Lagrangian** of the particle is therefore

$$L = \frac{1}{2}m \left(\dot{r}^2 + r^2 \sin^2 \alpha \dot{\phi}^2 \right) - mgr \cos \alpha.$$

Lagrange's equations are therefore

$$\begin{aligned}\frac{d}{dt}(m\dot{r}) - (mr \sin^2 \alpha \dot{\phi}^2) &= -mg \cos \alpha, \\ \frac{d}{dt}(mr^2 \sin^2 \alpha \dot{\phi}) - 0 &= 0.\end{aligned}$$

The second equation has the form $dp_\phi/dt = 0$, where $p_\phi = \partial L/\partial \dot{\phi}$, the momentum corresponding to ϕ . This is because $\partial L/\partial \phi = 0$, that is, ϕ is a **cyclic** coordinate. The momentum p_ϕ is the **angular momentum** of the particle about the vertical axis through O . ■

Example 2

Consider a ball thrown through the air. In the full three dimensions, the Lagrangian is

$$L = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - mgz.$$

There is no x or y dependence here, so both $\partial L/\partial \dot{x} = m\dot{x}$ and $\partial L/\partial \dot{y} = m\dot{y}$ are constant, as we well know. The fancy way of saying this is that conservation of $p_x \equiv m\dot{x}$ arises from spatial translation invariance in the x direction. The fact that the Lagrangian doesn't depend on x means that it doesn't matter if you throw the ball in one spot, or in another spot a mile down the road. The setup is independent of the x value. This independence leads to conservation of p_x .

Example 3

In spherical coordinates, consider a potential that depends only on r and θ . Our convention for spherical coordinates is that θ is the angle down from the north pole, and ϕ is the angle around the equator. The Lagrangian is

$$L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2) - V(r, \theta).$$

There is no ϕ dependence here, so $\partial L / \partial \dot{\phi} = mr^2 \sin^2 \theta \dot{\phi}$ is constant. Since $r \sin \theta$ is the distance from the z axis, and since $r \sin \theta \dot{\phi}$ is the speed in the tangential direction around the z axis, we see that our conserved quantity, $m(r \sin \theta)(r \sin \theta \dot{\phi})$, is the angular momentum around the z axis.

